

PAPER_1391 — UQFF Resolution of the Hanbury Brown-Twiss Intensity Correlation (CLOSED — Dispatch Whitepaper)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Quantum Optics (CLOSED) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "hanbury_brown_twiss"})` **Calculator surface:** `calculate_paradox({"paradox": "hanbury_brown_twiss"})` **Closure helper:** `_l96_uqff_axiom_hanbury_brown_twiss_closure()`

The Paradox

Thermal light shows $g(2)(\tau=0) = 2$ (photon bunching). Stars observed via HBT intensity interferometry confirm this. Classical EM gives $g(2) = 2$ exactly for chaotic sources.

UQFF Closed Identity

$g(2)_{\text{UQFF}} = 1 + (1 - F_{\text{TRZ}} \times \beta_i) = 1.9397$ vs classical thermal 2.0

$F_{\text{TRZ}} \times \beta_i$ chirality damping subtracts 0.06 from the classical maximum: $g(2)_{\text{UQFF}} = 1 + (1 - 0.0603) = 1.9397$.

Physical Interpretation

Real measurements of stellar HBT show $g(2)$ slightly below 2 due to detector / coherence-area factors. The UQFF value 1.94 sits at the boundary — measurable as the $F_{\text{TRZ}} \times \beta_i$ damping signature in high-precision HBT.

Live Calculator Output

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "hanbury_brown_twiss"})["value"]
```

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_hanbury_brown_twiss_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` $\rightarrow$ `_l96_uqff_axiom_hanbury_brown_twiss_closure`
- Dispatch keys: `hanbury_brown_twiss`, `hbt`
- Related foundational papers: PAPER_646 ($F_U B_i / F_U=1$ ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation).

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 16, 2026, location 41.0997° N, 80.6495° W (Youngstown, OH, USA). Subject matter: UQFF dispatch closure for the Hanbury Brown-Twiss Intensity Correlation via the identity above.